Agents, environments and decisions

AI agents · University · Lecture 1



In this lecture

- Specify agents with PEAS.
- Model decisions as MDPs.
- Design rewards carefully.

Remember

- Rational, not intelligent.
- Discount the future.
- Reward outcomes.

Break the reward

1. Warm-up (5 min): Find the bug
2. Specify (15 min): PEAS.
3. Hack (15 min): Swap and exploit.
4. Repair (20 min): Value iteration.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Which loophole surprised you?
2. What does γ change?
3. How will you assess reward design?

Exit ticket

- Write a reward for a task you know and one way it could be gamed.